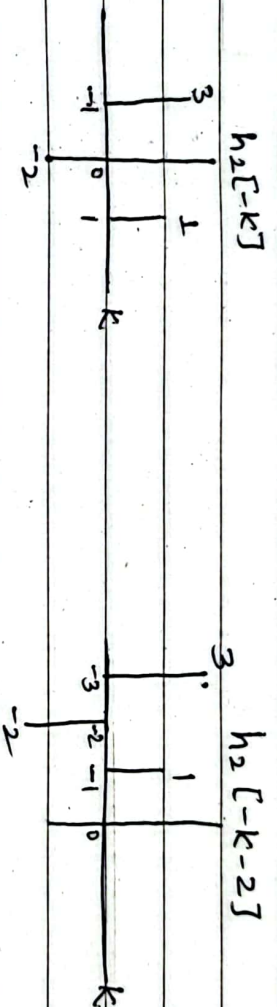
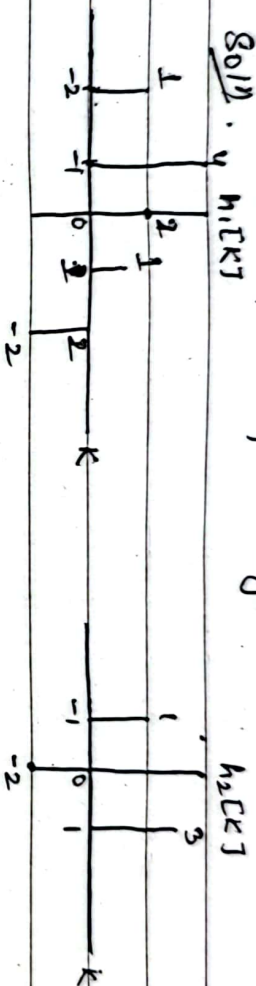


8. Given $h_1[n] = \{1, 4, 2, 1, -2\}$ and $h_2[n] = \{1, -2, 3\}$. Find the overall response $h[n]$ if the two systems are cascade interconnected. Also find the step response of this system.

Soln. $y[n] = h_1[k] * h_2[k]$



$$h[-2] = -2 + 4 = 2$$

$$h[-1] = 3 - 8 + 2 = -3$$

$$h[0] = 12 - 4 + 1 = 9$$

$$h[1] = 6 - 2 - 2 = 2$$

$$h[2] = 3 + 4 = 7$$

$$h[3] = -6$$

$$h[n] = \{2, -3, 9, 2, 7, -6\}$$

$$x[n] = \{1, 1, 1, 1, 1, 1\}$$

$$h[-n]$$

$$y[-2] = 2$$

$$y[-1] = -3 + 2 = -1$$

$$y[0] = 9 - 3 + 2 = 8$$

$$y[1] = 2 + 9 - 3 + 2 = 10$$

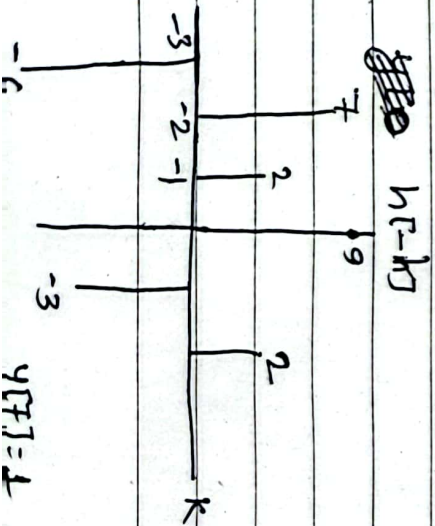
$$y[2] = 7 + 2 + 9 - 3 + 2 = 17$$

$$y[3] = 11$$

$$y[4] = -6 + 7 + 2 + 9 - 3 = 9$$

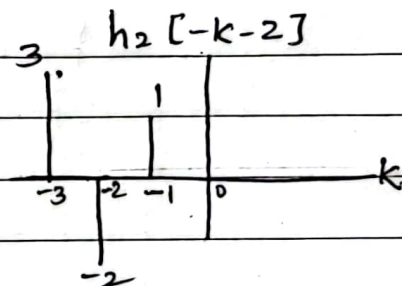
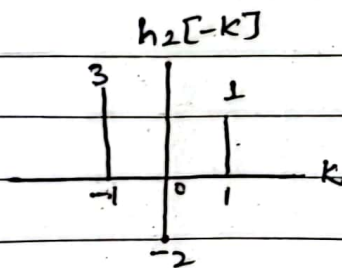
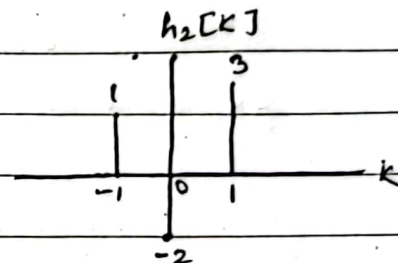
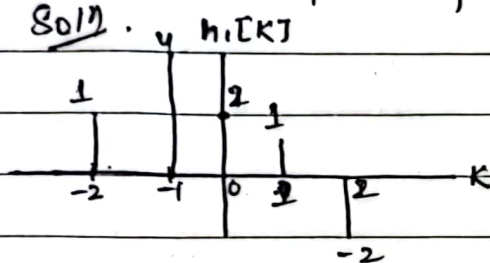
$$y[5] = -6 + 7 + 2 + 9 = 12$$

$$y[6] = -6 + 7 + 2 = 3$$



8. Given $h_1[n] = \{1, 4, 2, 1, -2\}$ and $h_2[n] = \{1, -2, 3\}$. Find the overall response $h[n]$ if the two systems are cascade interconnected. Also find the step response of this system.

Soln.



$$h[-2] = -2 + 4 = 2$$

$$h[-1] = 3 - 2 + 2 = -3$$

$$h[0] = 12 - 4 + 1 = 9$$

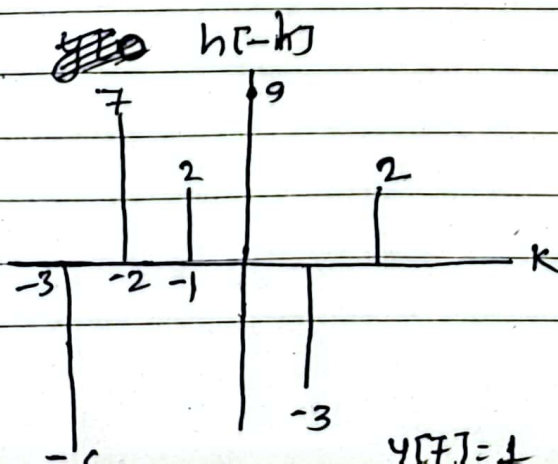
$$h[1] = 6 - 2 - 2 = 2$$

$$h[2] = 3 + 4 = 7$$

$$h[3] = -6$$

$$h[n] = \{2, -3, 9, 2, 7, -6\}$$

$$x[n] = \{1, 1, 1, 1, 1\}$$



$$y[-2] = 2$$

$$y[-1] = -3 + 2 = -1$$

$$y[0] = 9 - 3 + 2 = 8$$

$$y[1] = 2 + 9 - 3 + 2 = 10$$

$$y[2] = 7 + 2 + 9 - 3 + 2 = 17$$

$$y[3] = 11$$

$$y[4] = -6 + 7 + 2 + 9 - 3 = 9$$

$$y[5] = -6 + 7 + 2 + 9 = 12$$

$$y[6] = -6 + 7 + 2 = 3$$

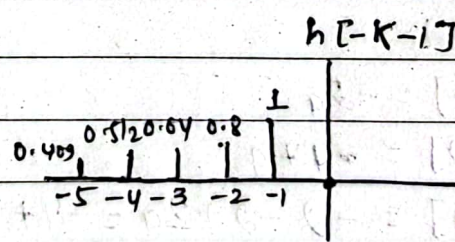
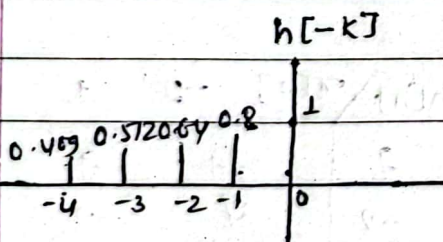
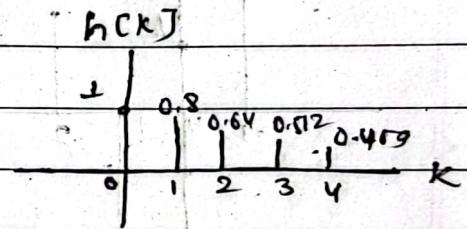
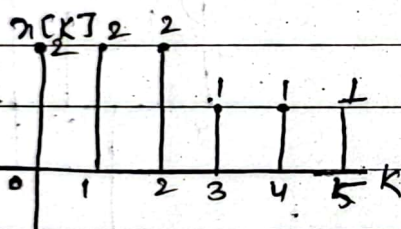
7b. $x[n] = 2u[n] - u[n-3] - u[n-6]; h[n] = 0.8^n u[n]$

Soln.

	0	1	2	3	4	5	6	7	8
$x[n]$	2	2	2	2	2	2	2	2	2
				1	1	1	1	1	1
							1	1	1

$x[n] = \{ \underset{\uparrow}{2}, 2, 2, 1, 1, 1, 0, 0, 0 \}$

$h[n] = \{ \underset{\uparrow}{1}, 0.8, 0.64, 0.512, 0.4096 \}$



$y[0] = 2$

$y[1] = 2 \times 0.8 + 2 \times 1 = 3.6$

$y[2] = 0.64 \times 2 + 0.8 \times 2 + 2 = 4.88$

$y[3] = 0.512 \times 2 + 0.64 \times 2 + 0.8 \times 2 + 1 = 4.904$

$y[4] = 0.4096 \times 2 + 0.512 \times 2 + 0.64 \times 2 + 0.8 + 1 = 4.922$

$y[5] = 0.4096 \times 2 + 0.512 \times 2 + 0.64 \times 1 + 0.8 + 1 = 4.282$

$y[6] = 0.4096 + 0.512 + 0.64 + 0.8 = 2.77$

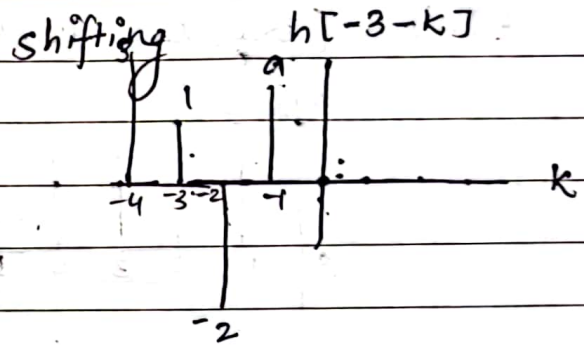
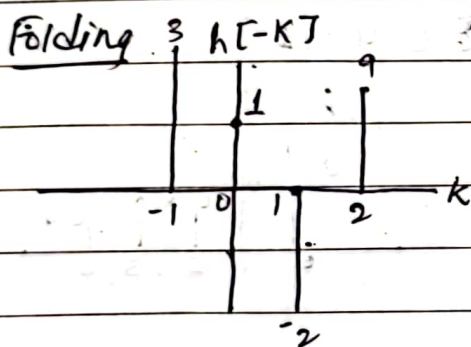
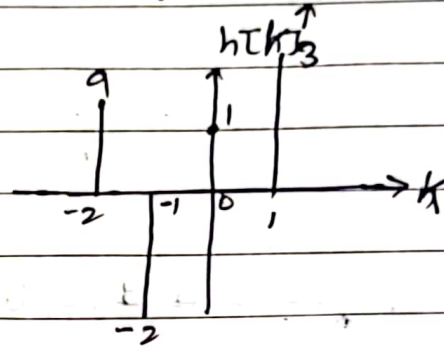
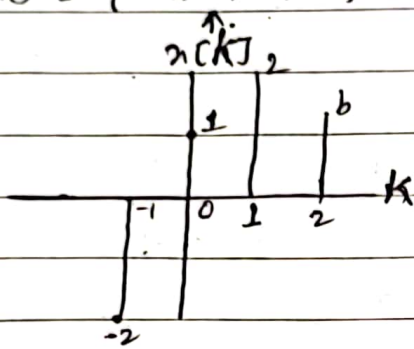
$y[7] = 0.4096 + 0.512 + 0.64 = 1.561$

$y[8] = 0.4096 + 0.512 = 0.921$

7. Determine $y[n]$.

a. $x[n] = \{-2, 1, 2, b\}$

$h[n] = \{a, -2, 1, 3\}$



$y[-3] = -2a$

$y[-2] = a + a$

$y[-1] = -2 - 2 + 2a = -4 + 2a$

$y[0] = -6 + 1 - 4 + ab$
 $= -9 + ab$

$y[1] = 3 + 2 - 2b$
 $= 5 - 2b$

$y[2] = 6 + b$

$y[3] = 3b$

$h[n]$ \ $x[n]$	-2	1	2	b
a	-2a	-a	2a	ab
-2	4	-2	-4	-2b
1	-2	1	2	b
3	-6	3	6	3b

e. The accumulator.

$$y[n] = y[n-1] + x[n]$$

Soln.

$$y_1[n] = y_1[n-1] + x_1[n]$$

$$y_2[n] = y_2[n-1] + x_2[n]$$

let $x[n] = a_1 x_1[n] + b_1 x_2[n]$

$$y[n] = a_1 y_1[n-1] + b_1 y_2[n-1] + a_1 x_1[n] + b_1 x_2[n]$$

And,

$$a_1 y_1[n] + b_1 y_2[n] = a_1 \{y_1[n-1] + x_1[n]\} + b_1 \{y_2[n-1] + x_2[n]\}$$

since $y[n] = a_1 y_1[n] + b_1 y_2[n]$ so linear.

For time invariance.

O/p due to delayed i/p.

$$y[n, k] = y[n-1-k] + x[n-k]$$

O/p due to delayed response

$$y[n-k] = y[n]$$

For causal.

$$y[n] = y[n-1] + x[n]$$

O/p depends on past and present value so causal.

For stability.

let $x[n]$ be bounded.

$$|x[n]| \leq M < \infty$$

$$|y[n]| = |y[n-1] + x[n]|$$

$$\leq |y[n-1] + M|$$

time dependent so unstable.

b. $y[n] = 2x[n^2]$

Solⁿ.

$$y_1[n] = 2x_1[n^2] \quad y_2[n] = 2x_2[n^2]$$

Let $x[n] = ax_1[n] + bx_2[n]$

we have $x[n^2] = ax_1[n^2] + bx_2[n^2]$

we have, $y[n] = 2x[n^2]$
 $= 2\{ax_1[n^2] + bx_2[n^2]\}$

$$ay_1[n] + by_2[n] = 2a \cdot x_1[n^2] + 2b \cdot x_2[n^2]$$

$$= 2\{ax_1[n^2] + bx_2[n^2]\}$$

Since $y[n] = ay_1[n] + by_2[n]$ so linear.

For time invariance.

o/p due to delayed i/p by k ~~is~~:

$$y[n, k] = 2x[(n-k)^2]$$

o/p due to delayed response by k

$$y[n-k] = 2x[(n-k)^2]$$

Here $y[n, k] \neq y[n-k]$ so time varying.

For causal

$$y[n] = 2x[n^2]$$

~~we~~ $y[1] = 2x[1]$

$$y[2] = 2x[4]$$

$$y[-2] = 2x[4]$$

Since o/p depends on present and future values so system is non causal.

For stability.

Let the i/p be bounded.

$$|x[n]| \leq M \quad n < \infty$$

$$|x[n^2]| \leq M \quad x.$$

$$|y[n]| = |2x[n^2]|$$

$$\leq |2 \cdot M|$$

$$\leq M_y < \infty \text{ so stable.}$$

Q: Check for the memory, linearity, time invariance, causality and stability of the following system.

a) $y[n] = 4n x[n]$

Soln

$$y_1[n] = 4n x_1[n]$$

$$y_2[n] = 4n x_2[n]$$

$$\text{Let } x[n] = a x_1[n] + b x_2[n]$$

$$\begin{aligned} \text{Here, } y[n] &= 4n x[n] \\ &= 4n \{a x_1[n] + b x_2[n]\} \end{aligned}$$

$$\begin{aligned} a y_1[n] + b y_2[n] &= a 4n x_1[n] + b 4n x_2[n] \\ &= 4n \{a x_1[n] + b x_2[n]\} \end{aligned}$$

So the system is linear.

For time invariance.

$$y[n-k] = 4(n-k) x[n-k]$$

$$y[n, k] = 4n x[n-k]$$

Since $y[n-k] \neq y[n, k]$ so time varying.

For causality.

$$y[n] = 4n x[n]$$

$$y[1] = 4 x[1]$$

↑
present value

Since o/p depend on present value so system is causal.

For stability.

Let the i/p be bounded.

$$|x[n]| \leq M \quad M < \infty$$

$$\begin{aligned} \text{then, } |y[n]| &= |4n x[n]| \\ &\leq 14n M \leq 14n1.M \\ &\leq M \quad \text{so unstable.} \end{aligned}$$

$$\begin{aligned} d \rightarrow \text{resolution} (\Delta) &= \frac{x_{\max} - x_{\min}}{L-1} \\ &= \frac{6+6}{1000-1} \\ &= 0.012 \text{ V} \end{aligned}$$

5. Determine the bit rate and the resolution in the sampling of a seismic signal with dynamic range of 0-1 volt if the sampling rate is $F_s = 20$ samples/s and we use 8 bit ADC. What is the maximum frequency?

Soln.

$$\text{Dynamic range} = 0.1 \text{ V} \Rightarrow x_{\max} - x_{\min}$$

$$\text{no of bits } (b) = 8$$

$$F_s = 20 \text{ samples/s}$$

Now, we have,

$$F_s = \frac{\text{bit rate}}{b}$$

$$\text{bit rate} = 20 \times 8 = 160 \text{ bits/sample} \checkmark$$

And,

$$\text{resolution} = \frac{\text{Dynamic range}}{2^b - 1}$$

$$= \frac{0.1}{2^8 - 1}$$

$$= 0.0003921 \text{ V} = 0.392 \text{ mV}$$

$$\text{Maximum frequency } (F_{\max}) = \frac{F_s}{2} = \frac{20}{2} = 10 \text{ samples/s}$$

$$f_1 = 3/10 \quad f_2 = 5/40$$

4. A digital communication link carries binary code words representing samples of an input signal $x_a(t) = 2\sin^2 300\pi t + 5\sin 500\pi t$. The link is operated at 10000 bits/sec and each input sample is quantized into 1024 different voltage levels.

Soln.

$$\text{Link speed (V)} = 10000 \text{ bits/s}$$

$$\text{No. of quantization level (L)} = 1024$$

Then, we know,

$$L = 2^b$$

$$\text{or, } 1024 = 2^b$$

$$b = 10 \text{ bits}$$

And,

$$a \rightarrow \text{Sampling frequency (F_s)} = \frac{V}{b} = \frac{10000}{10} = 1000 \text{ samples/s}$$

$$\text{folding frequency (F_{max})} = \frac{F_s}{2} = \frac{1000}{2} = 500 \text{ samples/s}$$

$$b \rightarrow \text{Nyquist rate (F_N)} = 2 F_{\max} \\ = 2 \times 500 = 1000 \text{ samples/s}$$

$$c \rightarrow x[n] = 2 \cdot \frac{1 - \cos 600\pi t}{2} + 5 \sin 500\pi t$$

$$x[n] = \frac{1 - \cos 2\pi \cdot 300 \times n}{1000} + 5 \sin \frac{2\pi \cdot 250 \times n}{1000}$$

$$= \frac{1 - \cos \left(2\pi \cdot \frac{3}{10} n \right)}{10} + 5 \sin \left(2\pi \cdot \frac{5}{20} n \right)$$

$$f_1 = \frac{3}{10}, \quad f_2 = \frac{5}{20}$$

6. Suppose that we sampled this signal at a rate of 150 samples/s what is the highest frequency that can be represented uniquely?
Soln.

$$F_s = 150 \text{ samples/s}$$

we know,

$$F_s = 2 F_{\max}$$

$$F_{\max} = \frac{150 \text{ sample/s}}{2} = 75 \text{ samples/s}$$

8. An analog signal $x_a(t) = 2 \sin(480\pi t) - 5 \sin(920\pi t)$ is sampled 800 times/s.
- Determine Nyquist rate.
 - Determine folding frequency.
 - What are the frequencies in radian in discrete time signal $x[n]$?
 - What is the analog signal $y_a(t)$ you can reconstruct?
- Soln.

$$F_s = 800 \text{ samples/s}$$

$$F_{\max} = 460 \text{ Hz}$$

Then,

$$F_N = 2 \times 460 \text{ Hz} \\ = 920 \text{ Hz}$$

$$\text{folding frequency} = \frac{F_s}{2} = \frac{800}{2} = 400 \text{ samples/s}$$

$$x[n] = 2 \sin\left(\frac{480\pi \times n}{800}\right) - 5 \sin\left(\frac{920\pi \times n}{800}\right)$$

$$= 2 \sin\left(2\pi \frac{24}{80} n\right) - 5 \sin\left(2\pi \frac{46}{80} n\right)$$

$$= 2 \sin\left(2\pi \left(\frac{3}{10}\right) n\right) - 5 \sin\left(2\pi \left(\frac{23}{40}\right) n\right) = -2 \sin\left(2\pi \frac{7}{10} n\right) - 5 \sin\left(2\pi \frac{17}{40} n\right)$$

6. Suppose that we sampled this signal at a rate of 150 samples/s what is the highest frequency that can be represented uniquely?

Soln.

$$F_s = 150 \text{ samples/s}$$

we know,

$$F_s = 2 F_{\max}$$

$$F_{\max} = \frac{150 \text{ sample/s}}{2} = 75 \text{ samples/s}$$

8. An analog signal $x_a(t) = 2 \sin(480\pi t) - 5 \sin(920\pi t)$ is sampled 800 times/s

- Determine Nyquist rate.
- Determine folding frequency.
- What are the frequencies in radian in discrete time signal $x[n]$?
- What is the analog signal $y_a(t)$ you can reconstruct?

Soln.

$$F_s = 800 \text{ samples/s}$$

$$F_{\max} = 460 \text{ Hz}$$

Then,

$$F_N = 2 \times 460 \text{ Hz} \\ = 920 \text{ Hz}$$

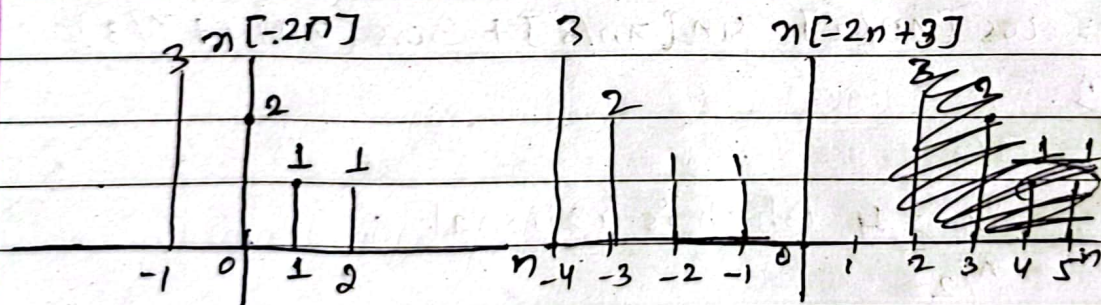
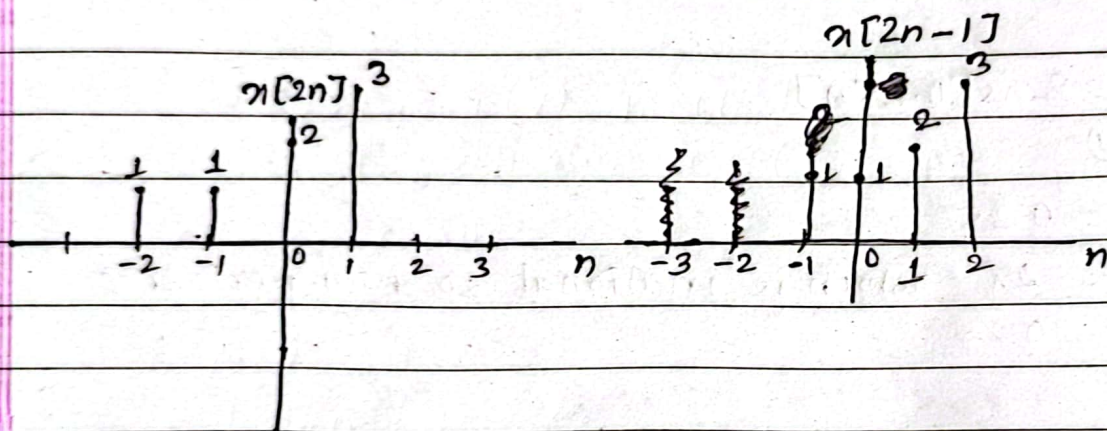
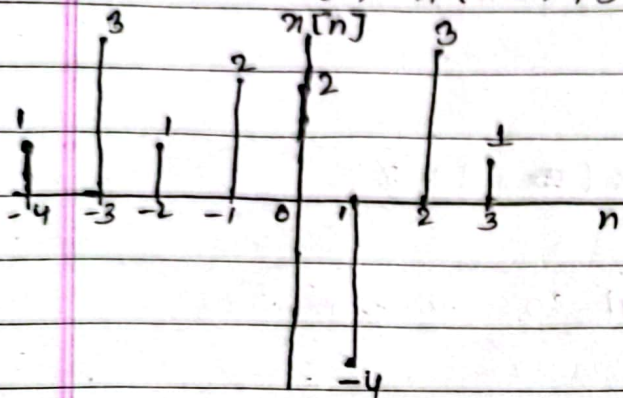
$$\text{folding frequency} = \frac{F_s}{2} = \frac{800}{2} = 400 \text{ samples/s}$$

$$x[n] = 2 \sin\left(480\pi \times \frac{n}{800}\right) - 5 \sin\left(920\pi \times \frac{n}{800}\right)$$

$$= 2 \sin\left(2\pi \frac{24}{80} n\right) - 5 \sin\left(2\pi \frac{46}{80} n\right)$$

$$= 2 \sin\left(2\pi \left(\frac{3}{10}\right) n\right) - 5 \sin\left(2\pi \left(\frac{23}{40}\right) n\right) = -2 \sin\left(2\pi \frac{1}{40} n\right) + 5 \sin\left(2\pi \frac{1}{40} n\right)$$

2. If $x[n] = \{1, 3, 1, 2, 2, -4, 3, 1\}$. Plot $x[n]$, $x[2n-1]$, $x[-2n+3]$

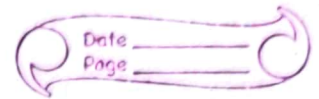


3. An analog ECG signal contains useful frequencies upto 100 Hz.

a. What is the Nyquist rate?

$$\text{Nyquist rate } (F_s) = 2 \times F_{\max} = 2 \times 100 \text{ Hz} \\ = 200 \text{ Hz}$$

Tutorial 1



1. Determine whether or not each of the signal is periodic. In case a signal is periodic, specify its fundamental period.

a) $x_a(t) = 3\cos(5\pi t + \pi/6)$

Soln.

Comparing it with $x(t) = A\cos(\omega t + \phi)$

$$\omega = 5\pi$$

$$T = \frac{2\pi}{5\pi} \text{ which is irrational so non-periodic.}$$

b) $x[n] = 3\cos[0.25n]$

Soln.

$$\omega = 0.25$$

$$T = \frac{2\pi}{0.25} \text{ which is irrational so non-periodic}$$

c) $x[n] = \cos[\pi n/2] - \sin[\pi n/8] + 3\cos(\pi n/4 + \pi/3)$

Soln.

$$\omega_0 = \pi/2$$

$$T_0 = \frac{2\pi}{\pi/2} = 4 \text{ which is rational}$$

$$\omega_1 = \pi/8$$

$$T_1 = 16$$

$$\omega_2 = \pi/4$$

$$T_2 = 8$$

So the signal is periodic.